THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 8 — MATHS · NCERT-TYPE UNIT TEST SET 3

Chapters 1–5: **A Square and a Cube, Power Play, A Story of Numbers, Quadrilaterals, Number Play** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. How many non-square numbers lie between 12^2 and 13^2? 1

(a) 23 **(b)** 12 **(c)** 24 **(d)** 25

2. How many zeros are there in one lakh written in figures? 1

(a) 5 **(b)** 4 **(c)** 6 **(d)** 7

3. Which of the following is NOT a correct Roman numeral? 1

(a) VX **(b)** CM **(c)** XC **(d)** CD

4. Which property holds for every parallelogram? 1

(a) The diagonals are equal **(b)** The diagonals are perpendicular
(c) All sides are equal **(d)** The diagonals bisect each other

5. The smallest digit that can replace * in 63*8 so that the number is divisible by 9 is: 1

(a) 8 **(b)** 1 **(c)** 5 **(d)** 2

6. The value of $20^2 - 19^2$ is: 1

(a) 40 **(b)** 39 **(c)** 38 **(d)** 20

7. The value of $\dfrac{1}{3^{-2}}$ is: 1

(a) -9 **(b)** 9 **(c)** 6 **(d)** $\frac{1}{9}$

8. The decimal value of 175_8 is: 1

(a) 127 **(b)** 130 **(c)** 125 **(d)** 123

9. The angles of a quadrilateral are in the ratio 2:3:4:6. The largest angle is: 1

(a) 96° **(b)** 156° **(c)** 132° **(d)** 144°

10. The sum of three odd numbers is: 1

(a) odd **(b)** a multiple of 3 **(c)** even **(d)** always prime

11. The triangular number T_{10} (sum of 1 to 10) is: 1

(a) 55 **(b)** 45 **(c)** 65 **(d)** 100

12. The value of $(-2)^5$ is: 1

(a) -32 **(b)** 10 **(c)** 32 **(d)** 37

13. The number 150 written in base 8 is: 1

(a) 227 **(b)** 226 **(c)** 622 **(d)** 225

14. In a parallelogram ABCD, $\angle A = (3x + 5)^\circ$ and $\angle C = (4x - 10)^\circ$. The value of x is: 1

(a) 14 **(b)** 20 **(c)** 15 **(d)** 16

15. Which of the following numbers is divisible by 11? 1

(a) 8241 **(b)** 8239 **(c)** 8238 **(d)** 8240

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Express 3^3 as the sum of 3 consecutive odd numbers. 2

17. The Earth is about 149600000 km from the Sun. Express it in standard form. 2

18. A Babylonian numeral is made of 3 sixties and 25 units, and a Maya numeral has 4 twenties and 11 units. Find their values. 2

19. Can a quadrilateral have angles 85°, 95°, 100° and 80°? Give a reason. 2

20. Take 13. Add 6, multiply by 3, then subtract 18. What is the result? How is it related to 13? 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. Show that the sum of 47 and its reverse is divisible by 11. 3

22. Show that $T_8 + T_9$ is a perfect square. 3

23. Express each as a single power of 3: 3

(i) $3^4 \times 3^{-6}$

(ii) $(3^2)^{-3}$

(iii) $3^5 \div 3^{-2}$.

24. Convert 214 to base 8 and check your answer by converting back to decimal. 3

OR

The angles of a quadrilateral are in the ratio 2:3:4:6. Find all the angles. 3

25. Check whether 73128 is divisible by 8 and by 11. 3

OR

Check whether $(26, 168, 170)$ is a Pythagorean triplet. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. (i) State two properties of the diagonals of a rhombus. 5

(ii) In rhombus ABCD the diagonals meet at O and $\angle BAD = 56^\circ$. Find $\angle OAB$, $\angle OBA$ and $\angle ABC$.

OR

Take 83 and its reverse 38. Find the difference and show that it is divisible by 9. Prove it in general for $10a + b$. 5

27. Verify that 4104 can be written as the sum of two cubes in two different ways: $2^3 + 16^3$ and $9^3 + 15^3$. 5

OR

Find x if

(i) $3^x \times 3^2 = 3^9$

(ii) $(3^x)^5 = 3^{35}$

(iii) $3^{x-4} = \dfrac{1}{3^1}$.

END OF QUESTION PAPER **Total: 50 Marks**

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